

Azure AI/ML Interview Questions (50 Questions)

Core Azure ML Services (Questions 1-10)

Q1: What is Azure Machine Learning and its key components?

Answer: Azure Machine Learning is a cloud-based platform for building, training, and deploying ML models.

Key Components:

- **Azure ML Studio:** Web-based IDE
- **Azure ML Workspace:** Central resource for ML projects
- **Azure ML Compute:** Training/inference compute clusters
- **Azure ML Datasets:** Data versioning and management
- **Azure ML Experiments:** Track training runs
- **Azure ML Pipelines:** ML workflow orchestration
- **Azure ML Models:** Model registry
- **Azure ML Endpoints:** Model deployment (real-time/batch)
- **Azure ML Designer:** Drag-and-drop ML pipeline builder
- **Automated ML:** AutoML capabilities

Workspace Hierarchy:

```
Subscription
├── Resource Group
│   └── ML Workspace
│       ├── Compute
│       ├── Datastores
│       ├── Datasets
│       ├── Experiments
│       ├── Models
│       └── Endpoints
```

Example:

```
from azureml.core import Workspace, Experiment, ScriptRunConfig

# Connect to workspace
ws = Workspace.from_config()

# Create experiment
experiment = Experiment(workspace=ws, name='training-experiment')

# Configure training
config = ScriptRunConfig(
    source_directory='./src',
    script='train.py',
```

```

    compute_target='cpu-cluster',
    environment=Environment.from_conda_specification(
        name='ml-env',
        file_path='conda.yml'
    )
)

# Submit run
run = experiment.submit(config)
run.wait_for_completion(show_output=True)

```

Q2: Explain Azure ML Compute types and when to use each.

Answer:

1. Compute Instance (Development):

- Personal cloud workstation
- Jupyter, VS Code, RStudio
- Always-on or scheduled shutdown
- Use: Development, prototyping

2. Compute Cluster (Training):

- Auto-scaling clusters
- CPU or GPU
- Scale to zero when idle
- Use: Training, batch inference

3. Inference Cluster (AKS):

- Azure Kubernetes Service
- Production workloads
- Auto-scaling
- Use: Real-time inference at scale

4. Attached Compute:

- Use existing VMs, Databricks, Synapse
- Bring your own compute
- Use: Integration with existing infrastructure

Comparison:

Compute Type	Purpose	Scaling	Cost
Instance	Dev/test	Fixed size	Always running
Cluster	Training	Auto-scale 0-N	Pay per use
AKS	Production inference	Auto-scale	Running cluster

Compute Type	Purpose	Scaling	Cost
Attached	Existing resources	Managed elsewhere	Varies

Example:

```

from azureml.core.compute import ComputeTarget, AmlCompute

# Create compute cluster
compute_config = AmlCompute.provisioning_configuration(
    vm_size='STANDARD_D3_V2',
    min_nodes=0,
    max_nodes=4,
    idle_seconds_before_scaledown=1800
)

compute = ComputeTarget.create(ws, 'cpu-cluster', compute_config)
compute.wait_for_completion(show_output=True)

```

Q3: What is Azure Automated ML and its capabilities?**Answer:**

Azure Automated ML (AutoML) automatically builds and tunes ML models.

Task Types:

- Classification (binary, multiclass)
- Regression
- Time-series forecasting
- Computer Vision (image classification, object detection)
- NLP (text classification, NER)

What AutoML Does:

- Feature engineering (scaling, encoding, imputation)
- Algorithm selection (tries multiple algorithms)
- Hyperparameter tuning
- Ensemble model creation
- Model explainability
- Guardrails (prevent overfitting)

Supported Algorithms:

- LightGBM, XGBoost, Random Forest
- Logistic Regression, Linear Regression
- Neural networks (TabNet, DNN)

Example:

```
from azureml.train.automl import AutoMLConfig

automl_config = AutoMLConfig(
    task='classification',
    primary_metric='accuracy',
    training_data=train_dataset,
    label_column_name='target',
    n_cross_validations=5,
    enable_early_stopping=True,
    experiment_timeout_hours=1,
    max_concurrent_iterations=4,
    featurization='auto',
    enable_voting_ensemble=True,
    enable_stack_ensemble=True
)

remote_run = experiment.submit(automl_config)
best_run, fitted_model = remote_run.get_output()
```

Q4: Explain Azure ML Pipelines and their components.

Answer:

Azure ML Pipelines are workflows for ML tasks.

Components:

1. Pipeline Steps:

- PythonScriptStep: Run Python scripts
- ParallelRunStep: Batch processing
- AutoMLStep: Automated ML
- DataTransferStep: Move data between datastores
- EstimatorStep: Train with estimators

2. Pipeline Data:

- PipelineData: Intermediate data between steps
- Datasets: Input data references

3. Pipeline Parameters:

- Runtime inputs (hyperparameters, file paths)

Benefits:

- Reusable workflows
- Parallel execution
- Scheduled runs
- CI/CD integration

- Cost optimization (caching)

Example:

```
from azureml.pipeline.core import Pipeline, PipelineData
from azureml.pipeline.steps import PythonScriptStep

# Intermediate data
processed_data = PipelineData('processed', datastore=ws.get_default_datastore())

# Step 1: Preprocess
preprocess_step = PythonScriptStep(
    name='preprocess',
    script_name='preprocess.py',
    arguments=['--output', processed_data],
    outputs=[processed_data],
    compute_target='cpu-cluster'
)

# Step 2: Train
train_step = PythonScriptStep(
    name='train',
    script_name='train.py',
    arguments=['--input', processed_data],
    inputs=[processed_data],
    compute_target='gpu-cluster'
)

# Create pipeline
pipeline = Pipeline(workspace=ws, steps=[preprocess_step, train_step])
pipeline_run = experiment.submit(pipeline)

# Publish for reuse
published_pipeline = pipeline.publish(name='ml-pipeline')

# Schedule
from azureml.pipeline.core.schedule import Schedule
Schedule.create(ws, name='daily-training',
               pipeline_id=published_pipeline.id,
               experiment_name='scheduled-training',
               recurrence=ScheduleRecurrence(frequency='Day', interval=1))
```

Q5-10: Core Azure ML Quick Reference

Q5: What are Azure ML Datasets?

Versioned data references. Tabular or File datasets. Track lineage. Monitor data drift. Registered in workspace.

Q6: Azure ML Model Registry features?

Central repository. Version control. Tagging and metadata. Model profiles. Deployment tracking. CI/CD integration.

Q7: What are Azure ML Environments?

Docker-based. Define Python packages, env variables. Curated (Microsoft-maintained) or custom. Reproducible training/inference.

Q8: Explain Azure ML MLflow integration.

Track experiments, log metrics, artifacts. Model registry. Deploy MLflow models. Cross-platform compatibility.

Q9: What is Azure ML Designer?

Drag-and-drop interface. No-code ML pipelines. Pre-built modules. Visual workflow. Export to Python SDK.

Q10: Azure ML Responsible AI Dashboard?

Model debugging, fairness assessment, error analysis, model interpretability, counterfactual what-if, causal analysis.

Azure Cognitive Services (Questions 11-20)

Q11: Compare Azure Computer Vision, Custom Vision, and Face API.

Answer:

Computer Vision:

- Pre-built image analysis
- OCR, object detection, image tagging
- Adult content detection
- Spatial analysis
- No training needed

Custom Vision:

- Train custom image models
- Classification and object detection
- Upload labeled images
- Export models (ONNX, TensorFlow)
- Use: Brand-specific recognition

Face API:

- Face detection and recognition
- Age, emotion, attributes
- Face verification (1:1)
- Face identification (1:N)
- Use: Security, attendance systems

Q12-20: Azure Cognitive Services Quick Reference

Q12: Azure Speech Services capabilities? Speech-to-text, text-to-speech, speech translation, speaker recognition. Custom models. Real-time and batch.

Q13: Language Understanding (LUIS) vs CLU? LUIS (legacy): Intent recognition. CLU (Conversational Language Understanding): Modern, better performance, multilingual.

Q14: Azure Translator features? 90+ languages. Document translation. Custom translation models. Real-time translation API.

Q15: Text Analytics capabilities? Sentiment analysis, key phrase extraction, NER, PII detection, language detection, opinion mining.

Q16: QnA Maker vs Question Answering? QnA Maker (legacy). Question Answering (part of Language Service): Better, integrated, improved ranking.

Q17: Azure Bot Service framework? Build chatbots. SDK (C#, Python, JavaScript). Channels: Teams, Slack, Web Chat. LUIS/CLU integration.

Q18: Form Recognizer capabilities? Extract text, tables, key-value pairs. Pre-built models: Invoice, Receipt, ID, Business Card. Custom models.

Q19: Anomaly Detector use cases? Time-series anomaly detection. Detect spikes, dips, trends. IoT monitoring, fraud detection.

Q20: Personalizer service? Reinforcement learning for personalization. Content recommendations. A/B testing. Learn from user actions.

Azure ML Advanced Topics (Questions 21-50)

Q21: What is Azure ML CLI v2?

Command-line interface. YAML-based job definitions. CI/CD integration. Simpler than v1.

Q22: Explain Azure ML online vs batch endpoints.

Online: Real-time, low latency, REST API. Batch: Large datasets, scheduled, async processing.

Q23: Azure ML managed online endpoints vs AKS?

Managed: Serverless, auto-scaling, simpler. AKS: More control, existing clusters, networking options.

Q24: What is Azure ML Model Monitoring?

Data drift detection. Model performance tracking. Alerts on degradation. Integration with App Insights.

Q25: Azure ML Feature Store (preview)?

Centralized feature management. Online and offline serving. Feature versioning. Point-in-time join.

Q26: Explain Azure ML parallel job.

Batch processing at scale. Process files in parallel. Auto-partitioning. Use: Scoring, ETL.

Q27: Azure ML sweep job?

Hyperparameter tuning. Grid, random, Bayesian search. Early termination policies.

Q28: What is Azure ML Registry?

Share models, environments, components across workspaces. Multi-workspace, multi-region.

Q29: Azure ML managed endpoints blue-green deployment?

Deploy new version without downtime. Test before switching traffic. Instant rollback.

Q30: Explain Azure ML component.

Reusable pipeline building blocks. Define interface (inputs, outputs). Share across teams.

Q31: Azure ML private link/VNet integration?

Secure workspace behind VNet. Private endpoints. No public internet access. Workspace isolation.

Q32: What are Azure ML curated environments?

Microsoft-maintained. TensorFlow, PyTorch, scikit-learn. Pre-tested, optimized. Regular updates.

Q33: Azure ML distributed training options?

Horovod, PyTorch DDP, TensorFlow MultiWorkerMirroredStrategy, DeepSpeed. Multi-node GPU training.

Q34: Explain Azure ML model profiling.

Predict resource requirements. Recommend instance type. Estimate latency. Cost optimization.

Q35: Azure ML cost estimation tools?

Cost calculator. Usage tracking. Budget alerts. Optimize: spot VMs, auto-shutdown, right-sizing.

Q36: What is Azure ML studio (classic)?

Legacy. Drag-and-drop interface. Being retired. Migrate to Azure ML Designer.

Q37: Azure ML integration with Azure Synapse?

Linked services. Train on Synapse Spark pools. Access Synapse data. Unified analytics + ML.

Q38: Azure ML integration with Databricks?

Attached compute. MLflow tracking. Deploy models. Use Databricks for data prep + training.

Q39: Explain Azure ML RunConfig vs ScriptRunConfig.

RunConfig (legacy). ScriptRunConfig: Simplified, recommended. Define training script + compute.

Q40: Azure ML model interpretation techniques?

SHAP, LIME, Permutation Feature Importance. Mimic explainer. Integrated with Responsible AI.

Q41: What is Azure ML model fairness assessment?

Fairness Metrics. Disparity analysis. Mitigation algorithms. Part of Responsible AI Dashboard.

Q42: Azure ML differential privacy?

SmartNoise toolkit. Privacy-preserving analytics. Add noise to protect individual data.

Q43: Explain Azure ML workspace diagnostics.

Health checks. Connectivity tests. Dependency validation. Troubleshooting tool.

Q44: Azure ML geo-distributed training?

Train across multiple regions. Data sovereignty compliance. Reduce data transfer.

Q45: What are Azure ML low-priority VMs?

Up to 80% cost savings. Preemptible. Use for non-critical training. Checkpointing recommended.

Q46: Azure ML model packaging formats?

ONNX, TensorFlow SavedModel, PyTorch, scikit-learn pickle. Supports most frameworks.

Q47: Explain Azure ML inference schema.

Define input/output format. Auto-generate Swagger. Client SDK generation. API documentation.

Q48: Azure ML model versioning strategies?

Semantic versioning. Tag latest, production. Immutable versions. Track deployment history.

Q49: What is Azure ML prompt flow (preview)?

Build LLM applications. Chain prompts. Test and evaluate. Deploy as endpoint.

Q50: Azure ML integration with GitHub?

GitHub Actions for CI/CD. Trigger training on commit. Automated model deployment. Version control for code + models.

Summary: Azure ML Services Comparison

Service	Purpose	Use Case
Azure ML	Full ML platform	Custom models, MLOps
Automated ML	AutoML	Quick model building

Service	Purpose	Use Case
Computer Vision	Image analysis	OCR, object detection
Custom Vision	Custom image models	Brand-specific recognition
Face API	Face recognition	Security, attendance
Speech Services	Speech processing	Transcription, TTS
Language Service	Text understanding	NER, sentiment
Translator	Translation	Multi-language content
Bot Service	Chatbots	Customer service
Form Recognizer	Document extraction	Invoice processing
Anomaly Detector	Anomaly detection	IoT monitoring
Personalizer	RL personalization	Content recommendations

Key Strengths:

- Enterprise integration (Active Directory, VNet)
- Strong MLOps capabilities
- Responsible AI toolkit
- Hybrid cloud support
- Azure ecosystem integration